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Chilling Accumulation Shapes Transcriptomic Responses to Warming Exposure During Grapevine Deacclimation

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ABSTRACT

During winter, grapevine (*Vitis vinifera*) bud dormancy and cold hardiness are regulated by complex interactions between chilling accumulation and warm temperature cues. However, the molecular mechanisms underlying physiological transitions during winter remain poorly understood. In this study, we performed time-series RNA-seq on “Cabernet Sauvignon” dormant buds with varying chilling accumulation, followed by warm temperature exposure under a controlled forcing condition. Using weighted gene co-expression network analysis, empirical modeling, and a novel calculation of molecular warm temperature response rate, we identified gene expression patterns responsive to warm temperature exposure alone, chilling alone, and their interaction. Genes responsive to warm temperature exposure under forcing conditions showed rapid, chilling-independent activation and were primarily associated with metabolism, environmental sensing, and auxin signaling. Chilling-responsive genes were enriched for functions of chromatin remodeling and heat shock protein pathways, which may indicate progressive cellular reprogramming under field conditions. Interaction-responsive genes, including those involved in ABA/auxin metabolism and cell wall modification, may function in both dormancy progression and deacclimation. These findings provide a transcriptomic framework suggesting how chilling and temperature synergistically regulate dormancy transitions in grapevine, thereby enhancing the understanding of temperature sensing and response and the chilling-mediated dormancy progression underlying grapevine dormant season physiology.

1 | Introduction

Driven largely by human activities, climate change is marked by persistent shifts in temperature and precipitation patterns, alongside a rise in the frequency and intensity of extreme weather events (Santos et al. 2020). These changes are disrupting global forest distributions and impacting the long-term sustainability of agricultural systems worldwide (Raza et al. 2019; Salama et al. 2021). Perennial crops are especially vulnerable to these climatic stresses due to their long life cycle and high re-establishment costs, which limit the feasibility of rapid adaptation strategies such as cultivar replacement.

In the growing season, heat stress and drought stress impair physiological function and metabolite biosynthesis, which ultimately lowers the quality of fruits and derived products like wine and cider (AghaKouchak et al. 2020; van Leeuwen et al. 2024). During the dormant season, warmer average winter temperatures can increase the risk of cold damage by disturbing the process of cold acclimation, particularly when followed by extreme freezing events (Kovaleski 2024). This is particularly true for the cultivation of domesticated grapevines (*Vitis vinifera*) in cool or cold climate regions in North America and Eurasia, where the lowest temperatures occasionally drop below their maximum cold hardiness (Zabadal et al. 2007;

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Wang and Londo 2025). In addition, more frequent spring frost damage, a global climatic concern, can be associated with high crop loss in grapevine as the damage often occurs in primary buds, which are the most fruitful (Wang and Dami 2020; Persico et al. 2021; Poni et al. 2022). A deeper understanding of grapevine dormant season physiology is essential for safeguarding the sustainability of current production and supporting the continued expansion of the grape and wine industries under climate change.

It is widely believed that the vulnerability of grapevines during the dormant season is closely tied to the coordinated regulation of dormancy and cold hardiness. Grapevines and other perennial species undergo specific physiological changes to enter and maintain dormancy throughout winter, while simultaneously developing cold hardiness to withstand subfreezing temperatures. Dormancy, characterized by the cessation of visible growth, is generally classified into three categories: paradormancy, endodormancy, and ecodormancy (Lang 1987). Paradormancy refers to the suppression of growth caused by physiological factors, primarily auxin-mediated apical dominance, during the growing season, such as the inhibition of lateral bud outgrowth by the shoot apex (Fan et al. 2023). At the end of the growing season, grapevine buds transition from paradormancy to endodormancy in response to shortening daylength and decreasing temperatures (Fennell 2004; Grant et al. 2013). During endodormancy, growth is suppressed and buds have reduced sensitivity to environmental cues, a mechanism that protects them from responding prematurely to fluctuating conditions (Yang et al. 2021). Concurrent with the onset of endodormancy, grapevines transition from a cold-sensitive to a cold-hardy state through the process of cold acclimation (Gusta et al. 2005). To resume growth in spring, grapevines must transition from endodormancy to ecodormancy, where environmental sensitivity is restored but growth remains suppressed due to ambient low temperatures. When ecodormant buds are exposed to warm temperatures, they begin to lose cold hardiness through a temperature-dependent process known as deacclimation (Pagter and Arora 2013; Wisniewski et al. 2018; Kovaleski et al. 2018; Londo and Kovaleski 2025). The transition from endodormancy to ecodormancy is thought to be triggered by the accumulation of chilling, the cumulative exposure to temperatures within a specific range during the dormant season (Dokoozlian 1999; Campoy et al. 2011; Wang et al. 2020; Kovaleski 2022). Various models have been developed to quantify chilling accumulation, with widely used approaches such as the North Carolina model (Shaltout and Unrath 1983), the Utah model (Richardson et al. 1974), and the Dynamic Model (Fishman et al. 1987). These models generally consider temperatures between 0°C and 15°C as effective for chilling, though they differ in their optimal temperature ranges and in how they weigh the efficiency of chilling accumulation. A new model expanded the range of chilling-effective temperatures to include freezing conditions and incorporated temperature fluctuations as an additional factor, resulting in improved chilling estimates that better correlate with physiological changes (North et al. 2024).

Because chilling accumulation and temperature exposure jointly regulate dormant-season physiology, previous studies have begun to examine the molecular processes underlying

cold hardiness dynamics and dormancy transitions. As part of the cold hardiness dynamics, cold acclimation is typically divided into three sequential phases, initiated by shortening daylength, followed by exposure to low above-freezing temperatures (4°C–10°C), and finally by subfreezing temperatures (<0°C) (Fennell 2004; Zabadal et al. 2007; Grant et al. 2013). Daily temperature fluctuations are also critical for initiating cold acclimation (Wang, Kovaleski, and Londo 2024; North et al. 2024). During cold acclimation, cold hardiness is progressively enhanced through a complex interplay of genetic regulation, metabolic adjustments, and physiological adaptations. Key genetic responses include the activation of hormone signaling pathways, such as abscisic acid (ABA), and in many species, the inducers of CBF Expression–C-Repeat Binding Factor/DRE Binding Factor–Cold Regulated Genes (*ICE-CBF/DREB1-COR*) cascade are activated, which together promote the production of functional proteins and metabolites. These molecular responses drive downstream physiological and structural changes, including cell dehydration, membrane lipid remodeling, and increased supercooling capacity, all of which contribute to enhanced cold hardiness (Noriega and Pérez 2017; Londo et al. 2018; Liu et al. 2019; Chai et al. 2019; Ritonga and Chen 2020; Ouyang et al. 2021; Hwarari et al. 2022; Noriega et al. 2024; Ralser et al. 2024; Konecny et al. 2026).

In contrast to cold acclimation, deacclimation represents the loss of cold hardiness under warm conditions and is strongly influenced by previous chilling exposure. As chilling exposure increases, grapevine deacclimation rates (k_{deacc}) increase; however, substantial variation in chilling requirements and maximum deacclimation rates has been observed among *Vitis* cultivars with diverse genetic backgrounds (Londo and Johnson 2014; Londo and Kovaleski 2025). At the molecular level, the deacclimation of grapevine has been found to be correlated with a plant hormone metabolism and signaling network that involves ABA degradation and signaling, ethylene biosynthesis and signaling, jasmonate and gibberellin biosynthesis, auxin signaling and cytokinin degradation along with the activation of growth-related pathways that include cell wall modification, ribosome biogenesis and photosynthesis protein biosynthesis (Kovaleski and Londo 2019; De Rosa et al. 2025; Wang et al. 2025).

Although these studies have established chilling as a major physiological regulator of dormancy transition and deacclimation capacity, the molecular mechanisms by which buds perceive and respond to chilling remain poorly resolved. It is widely believed that the chilling-mediated transition from endodormancy to ecodormancy is a consequence of plant hormone interactions and their downstream signaling (Atkinson et al. 2013; Brunner et al. 2014; Beauvieux et al. 2018; Yang et al. 2021; Penfield et al. 2021). The onset of endodormancy is characterized by elevated levels of ABA and reduced levels of gibberellins (GAs), whereas endodormancy release is associated with an increase in gibberellin concentrations (Yang et al. 2021). In both pear and grapevine buds, chilling-mediated dormancy release has been linked to a decline in ABA levels, which corresponds with downregulation of ABA biosynthetic genes and upregulation of ABA catabolism genes (Zheng et al. 2015; Tuan et al. 2017). Exogenous applications of various plant hormones or their analogs have been shown to disrupt the natural

deacclimation process in plants (Jacobsen et al. 2013; Zheng, Kwame Acheampong, et al. 2018; Kovaleski and Londo 2019; Yuxi et al. 2021; Wang et al. 2022, 2025). A group of MADS-box transcription factors, known as DORMANCY-ASSOCIATED MADS-box genes (DAMs), has been noted for their potential regulation of dormancy transition under chilling accumulation. In *Prunus persica*, six tandemly arranged DAM genes located at the end of chromosome 1 were identified, and a deletion at this locus was associated with the “evergrowing” phenotype (Bielenberg et al. 2008). The expression of all six DAM genes is highly correlated with chilling accumulation (Li et al. 2009; Yamane et al. 2011; Yu et al. 2020). In addition, chilling during the dormant season was also found to induce DNA methylation changes, particularly in the CHH context, in perennial crops like apple (*Malus domestica*) and sweet cherry (*Prunus avium*), regulating gene expression for dormancy release and growth preparation (Kumar et al. 2016; Rothkegel et al. 2020). While several physiological studies have explored the relationship between chilling accumulation and dormancy progression, the underlying molecular mechanisms remain poorly understood.

To address this knowledge gap, we conducted an experiment involving five field collections of grapevine buds across the dormant season, each representing a different level of field-accumulated chilling. Following each collection, buds were subjected to a controlled forcing assay simulating growth-permissive conditions to capture dynamic transcriptomic responses to warm temperature exposure and field chilling. Our objectives were to: (1) investigate the molecular basis of the dormant bud response to warm temperature; (2) determine whether these responses differ according to field chilling; and (3) identify the portion of the transcriptome that is responsive to field chilling only, independent of warming. We hypothesized that field chilling alters hormone signaling and transcriptional readiness in dormant buds. These changes may reactivate temperature sensitivity and enable deacclimation-associated growth-related pathways upon exposure to warm conditions. This research provides critical insights into the molecular transition from endodormancy to ecodormancy and is essential for improving climate resilience in perennial crop systems.

2 | Materials and Methods

2.1 | Plant Materials and Field Sampling

Field-grown *Vitis vinifera* “Cabernet Sauvignon” grapevines were used in this study. The experimental vines were commercially cultivated at Ravines Wine Cellars in Geneva, NY (42.845°N, 77.004°W). Vines were grafted onto 3309C rootstock, trained to a vertical shoot-positioned (VSP) system, and maintained under standard commercial vineyard management practices, including routine pruning, canopy management, and pest and disease control. The experimental design and detailed sample collection workflow are outlined in Figure 1A. Dormant grapevine canes were collected at five timepoints during the 2021–2022 dormant season: November 3 (Collection 1), November 16 (Collection 2), December 1 (Collection 3), December 14, 2021 (Collection 4), and February 20, 2022 (Collection 5) to represent approximately equal chilling intervals spanning the transition from endodormancy to ecodormancy. For each field collection, a

total of ten 10-bud canes were collected from field-grown grapevines. Although there is a larger gap in calendar time between Collections 4 and 5, this interval reflects the slower rate of chill accumulation during mid-winter (Figure 1). In this study, “field chilling” refers to chilling units accumulated in situ by the grapevines under commercial vineyard conditions before cane collection. Thus, field-grown vines were the recipients of chilling accumulation, while the excised canes and buds collected from these vines were used as the experimental material for subsequent deacclimation-forcing assays.

2.2 | Deacclimation-Forcing Assays

After each field collection, canes were pooled regardless of source vine or individual cane and then chopped into single-bud cuttings. The single-bud cuttings were incubated without supplemental light with cut ends in water and held at 22°C in a growth chamber. Such a growth-permissive forcing assay has been used to determine the deacclimation potential and the dormancy status of grapevine (Londo and Johnson 2014; Wang, Kovaleski, and Londo 2024; Londo and Kovaleski 2025). Bud cold hardiness was assessed at 0 h (field condition), 24 h (1 day), 72 h (3 days), 168 h (7 days), and 288 h (12 day) after the start of forcing if the buds remained unbroken. The examined buds were from the same pooled field collection and were sampled at the same time point during the forcing assays. Therefore, buds used for cold hardiness measurements were not necessarily from the same individual cane or vine but from the same pooled cane collection exposed to the same field chilling and forcing conditions. Deacclimation rate (k_{deacc} , in °C·h⁻¹ or °Cd⁻¹), the coefficient of the linear regression between bud cold hardiness and time under forcing conditions, was determined for each deacclimation assay conducted on the samples from each field collection (Kovaleski et al. 2018).

2.3 | Cold Hardiness Measurement

Bud cold hardiness was measured using differential thermal analysis (DTA) following standard protocols (Mills et al. 2006; Londo et al. 2023). Buds were excised along with surrounding cane tissue from single-node cuttings and placed into thermoelectric modules. Samples were then subjected to controlled freezing in a Tenney T2C environmental chamber (Tenney Environmental), programmed to cool at a rate of 4°C·h⁻¹ from 0°C to –50°C. The release of heat during tissue freezing, a phenomenon known as the low temperature exotherm (LTE), was detected and recorded via a Keithley 2700 data logger (Tektronix). The temperature at which the LTE occurs corresponds to the cold hardiness of the grapevine bud (Pierquet and Stushnoff 1980; Londo et al. 2023). Cold hardiness was measured at each timepoint of the deacclimation assays under growth chamber conditions using five biological replicates per field collection. Each biological replicate represents one individual bud sampled from the pooled single-bud cuttings, rather than from an individual vine or cane. In addition, field cold hardiness of “Cabernet Sauvignon” grapevines from the same vineyard was continuously monitored throughout the dormant season to serve as a reference for evaluating deacclimation responses under controlled conditions.

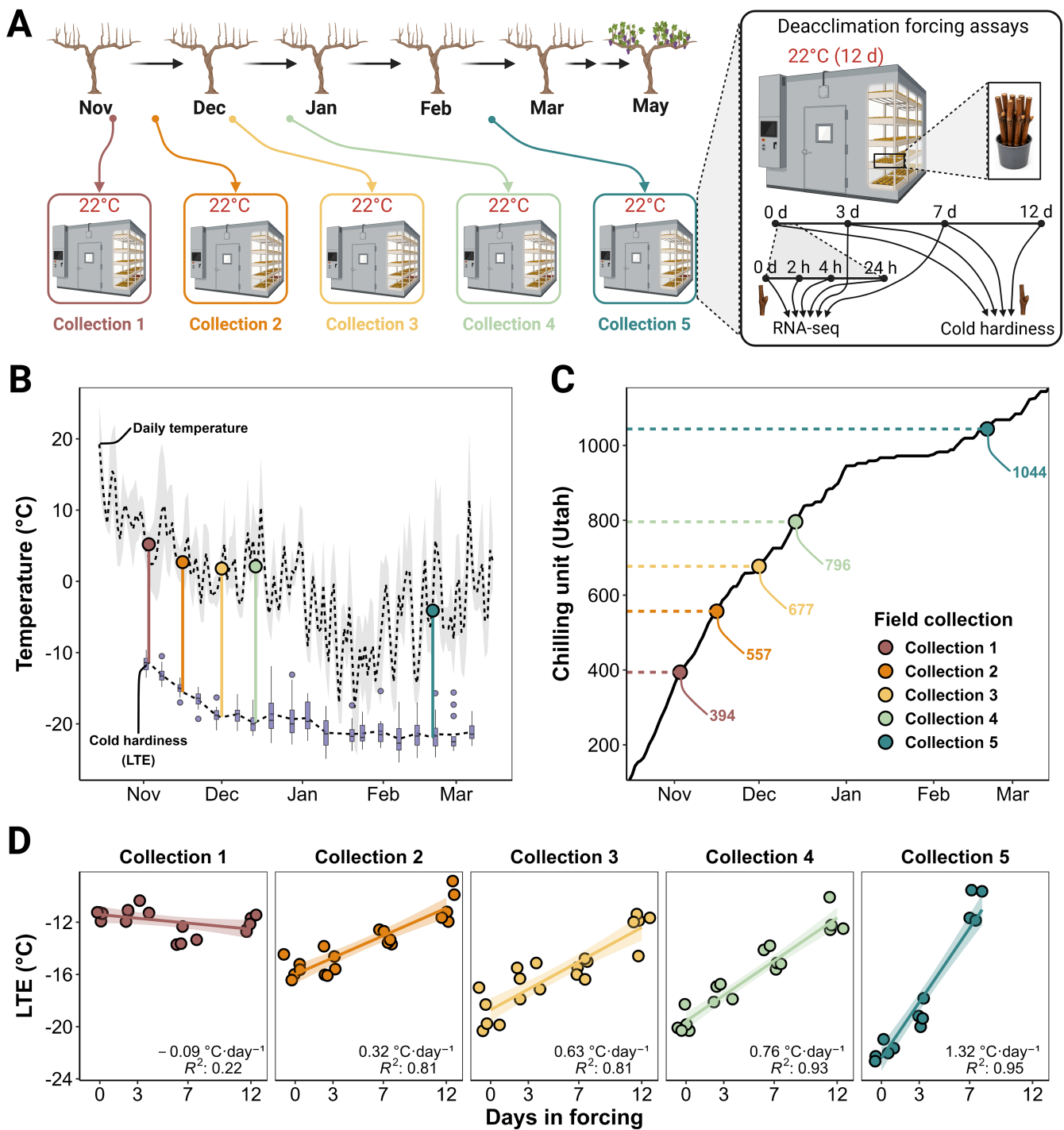


FIGURE 1 | Experimental design, sample collection, data collection, and deacclimation assays. (A) Schematic overview of the experimental design and sampling timeline. The inset illustrates the timing of sample collection within each deacclimation forcing assay for RNA-seq analysis and bud cold hardness determination; (B) Daily ambient temperature throughout the 2021–2022 dormant season, along with corresponding field temperature and bud cold hardness at each sampling timepoint; (C) Chilling unit accumulation based on the NC model and corresponding chilling units at each sampling timepoint; (D) Bud cold hardness in the forcing assays and the computed k_{deacc} . The figure was assembled in and the illustrations were obtained from BioRender (<https://www.biorender.com/>).

2.4 | RNA-seq Data Collection

During each deacclimation assay, bud samples were collected for transcriptomic analysis at 0 h (field condition), 2, 4, 24, 72, and 168 h after the start of deacclimation forcing assays. These RNA-seq sampling timepoints partially overlapped with, but were not identical to, the timepoints used for cold hardness

measurements. For each RNA-seq timepoint, three to four biological replicates were prepared, with each biological replicate consisting of five pooled buds. Total RNA was extracted, and library construction was performed using the Lexogen QuantSeq 3'mRNA-Seq Library Prep Kit (Lexogen), with technical support provided by the Cornell University Institute of Genomic Diversity (Ithaca, NY, USA). Sequencing was carried out at

the Cornell University Institute of Biotechnology (Ithaca, NY, USA) using the Illumina NextSeq500 platform (Illumina Inc.). Libraries were sequenced with a read length of 85bp, and each library was sequenced in triplicate to ensure technical reproducibility.

2.5 | RNA-Seq Data Analysis

RNA-seq data were analyzed following a previously generated pipeline specifically optimized for grapevine (Wang et al. 2022; Wang, Kovalski, and Londo 2024). Technical replicates from the same biological library were pooled prior to downstream analysis, which resulted in one count profile per biological replicate. Standardized library QC (FastQC) and trimming (BBduk) were conducted for quality control of the libraries, and transcript alignment was conducted in STAR using *V. vinifera* 12X.v2 genome and VCost.v3 annotation (Dobin et al. 2013; Canaguier et al. 2017). DESeq2 was used to identify differentially expressed genes (Love et al. 2014). Principal component analysis (PCA) and weighted gene co-expression network analysis (WGCNA) were used for the characterization of the transcriptome and the identification of gene co-expression modules (Langfelder and Horvath 2008). After the identification of target gene co-expression modules that revealed specific behavior under field or forcing conditions, we applied specific contrasting or correlation-based filtering to identify target genes.

To identify the genes that responded to the warm temperature exposure in the forcing assays (co-expression modules “turquoise” and “blue”), we generated a molecular temperature model and calculated a molecular temperature response rate (k_T) based on the module eigengene (ME) expression patterns of the “turquoise” and “blue” co-expression modules. This model is based on the assumption that the rate of expression change in major warm-temperature-responsive gene modules reflects the bud’s molecular response to growth-permissive warm temperature exposure. The calculation was performed as follows:

$$k_T = \left(\frac{\Delta E_{24h}}{\Delta E_{total}} \right) \cdot \frac{\Delta T}{24} \quad (1)$$

where k_T represents the molecular response rate to temperature change, expressed in $^{\circ}\text{C h}^{-1}$; ΔE_{24h} is the change in eigengene expression in the Collection 5 samples during the first 24 h of forcing; ΔE_{total} is the total change in eigengene expression in the Collection 5 samples over the entire assay, calculated relative to the mean eigengene expression values on days 3 and 7 from all the samples; ΔT is the difference in temperature between the Collection 5 field condition and the constant 22°C growth chamber environment. The resulting k_T describes the rate at which gene expression responds to temperature change between the field condition and the controlled environment. The k_T was calculated separately for the WGCNA modules “turquoise” and “blue,” representing up-regulated and downregulated genes under forcing conditions, respectively.

Assuming that temperature-responsive genes in grapevine respond linearly to changes in temperature, the molecular

temperature that reveals the current expression at a given timepoint can be modeled using the molecular temperature-response rate k_T . The calculated molecular temperature T_{calc} is given by:

$$T_{calc} = T_0 + X \cdot k_T \quad (2)$$

where T_0 is the baseline temperature under field condition, X is the time under the forcing condition, and k_T is the molecular response rate to temperature change. The actual molecular temperature at each timepoint, $T_{molecular}$, is constrained by a temperature limit of 22°C in the forcing assay, and is defined as:

$$T_{molecular} = \begin{cases} 22^{\circ}\text{C}, & T_{calc} \geq 22^{\circ}\text{C} \\ T_{calc}, & T_{calc} < 22^{\circ}\text{C} \end{cases} \quad (3)$$

This formulation ensures that gene expression does not exceed its empirically observed target value while still capturing the linear response to temperature-driven deacclimation. Based on this formulation, the $T_{molecular}$ was calculated for each field collection and growth chamber data collection timepoints and was used as the only numeric factor in a gene expression model in DESeq2 to identify these temperature responsive genes. Only the genes with a $p_{adj} \leq 1e-10$ in such a model were identified as significant temperature responsive genes.

To identify genes that responded to field chilling (module “black”) or to the interaction between chilling and warm temperature exposure (modules “brown” and “yellow”), we constructed a gene expression model using the formula: *design* = ~chilling + chilling × time_in_forcing. For genes associated with both chilling and warm temperature exposure response (modules “brown” and “yellow”), significance was determined by a $p_{adj} \leq 1e-5$ for both the chilling term and the interaction term. These genes were classified as significant temperature-responsive genes.

For genes that responded to chilling but did not change over time during the experiment (module “black”), significance was defined as $p_{adj} \leq 1e-5$ for the chilling term and $p_{adj} > 1e-3$ for the interaction term, which specifies a chilling-specific but warm temperature-independent expression pattern.

In addition, based on the computed k_{deacc} and the measured initial cold hardiness, we also estimated the cold hardiness of the buds at each timepoint during the forcing assays. This calculation allowed us to assess the correlation between bud cold hardiness dynamics and the MEs of modules “brown” and “yellow” during deacclimation. The calculated cold hardiness (CH_{cal}) at any timepoint is defined as:

$$CH_{cal} = CH_0 + X \cdot k_{deacc} \quad (4)$$

where CH_{cal} ($^{\circ}\text{C}$) is the calculated cold hardiness, CH_0 ($^{\circ}\text{C}$) is the initial cold hardiness under field conditions (0h in the forcing assays), X is the time under the forcing condition (in hours), k_{deacc} is the calculated deacclimation rate in $^{\circ}\text{C h}^{-1}$ to facilitate the calculation of the cold hardiness within the first 4 h.

2.6 | Pathway Enrichment Analysis and Other Examinations

Enriched pathways among the different sets of identified target genes were detected through over-representation analysis using VitisNet 12X functional pathways (Grimplet et al. 2009). The analysis was conducted following a standard pathway enrichment analysis pipeline for grapevine (Wang et al. 2022). VitisNet 12X-derived pathway annotations were reformatted into a Gene Matrix Transposed (.gmt) file and used for over-representation analysis. The files and code required for this analysis were previously released in our grapevine 3'mRNA-Seq analysis pipeline repository (<https://github.com/imbaterry11/3-RNA-seq-analysis-pipeline-for-grape>). Significantly enriched pathways at $p_{\text{adj}} \leq 0.005$ were examined for correlations with known biological functions, and hub genes were determined based on the synergy of statistical significance and biological functionality. In addition, we also manually examined individual genes or the genes involved in pathways that were previously reported to be functional in grapevine cold hardiness. To simplify interpretation and visualization, expression values of all gene copies encoding the same protein were summed in each figure, unless the discussion focused on individual genes. Numbers in parentheses indicate the number of gene copies included in each sum.

3 | Results

3.1 | Grapevine Cold Hardiness Dynamics Under Field Condition

We conducted a time-series experiment to assess the physiological and transcriptomic responses of *Vitis vinifera* “Cabernet Sauvignon” grapevines collected from the field during the 2021–2022 dormant season. Cane samples were collected on November 3 (Collection 1), November 16 (Collection 2), December 1 (Collection 3), December 14, 2021 (Collection 4), and February 20, 2022 (Collection 5) (Figure 1A). The ambient temperatures at the time of collection were 5.2°C, 2.7°C, 1.8°C, 2.1°C, and −4.1°C, respectively (Figure 1B). The field cold hardiness of the grapevine buds at each field collection date was measured as −11.4°C, −15.5°C, −18.8°C, −19.8°C, and −21.9°C, respectively (Figure 1B). To estimate chilling accumulation in the field, we applied the (Utah) chilling unit model (Shaltout and Unrath 1983), which resulted in 394, 557, 677, 796, and 1044 chilling units for Collections 1 through 5, respectively (Figure 1C).

3.2 | Diverged Deacclimation Under Growth-Permissive Condition

To examine the deacclimation response (k_{deacc}), cane samples collected from the field were chopped into single-bud cuttings, which were placed with cut ends in cups of DI water and incubated under a growth-permissive forcing condition. Here, k_{deacc} represents the rate of cold hardiness loss under forcing conditions, with higher values indicating faster deacclimation and greater bud responsiveness to warm forcing. All sample collections exhibited significant deacclimation responses (Figure 1D)

except those from Collection 1. Instead, a negative k_{deacc} slope was observed for samples in Collection 1, and the regression between cold hardiness and time in forcing showed minimal fit (lowest R^2 ; Figure 1D). Samples from Collections 2 through 5 showed increasing deacclimation capacity, with all corresponding regressions showing R^2 values above 0.8. The calculated k_{deacc} steadily increased with chilling accumulation, ranging from 0.42°C d^{−1} in Collection 2 to 1.42°C d^{−1} in Collection 5.

3.3 | Transcriptomic Characterization of Bud Samples During Controlled Deacclimation

In parallel with each deacclimation assay, replicate buds were analyzed using RNA-seq to characterize their transcriptomic response to forcing temperatures and varying levels of chilling accumulation. A total of 95 RNA-seq libraries were generated from the five deacclimation assays. All libraries passed FastQC per-base quality standards, with an average of 5.2 million reads per library and a mean unique mapping rate of 81.2%. One library was identified as an outlier based on PCA and was excluded from downstream analyses (Figure S1). After filtering out low-count genes (defined as fewer than 10 transcripts per library), the final dataset consisted of 94 libraries and 10,428 genes, representing 24.6% of the total annotated grapevine transcriptome.

To characterize transcriptomic trends, PCA and WGCNA were applied to the filtered expression matrix. PC1 and PC2 accounted for 27.2% and 11.2% of the total variance, respectively, differentiating libraries based on field chilling history and response to forcing conditions (Figure 2A). Transcriptomic profiles within each field collection were largely similar when examining time-points collected in the first 4 h of exposure to forcing conditions (0, 2, and 4 h). In contrast, by 24 h, libraries from all field collections shifted substantially toward the upper left quadrant of the PC1/PC2 plot, suggesting a transcriptomic transition between the within-a-day (0 to 4 h) and after-a-day (≥ 24 h) sample points, regardless of initial field conditions (Figure 2A). Libraries from Collections 1 to 4 exhibited limited transcriptomic change after 24 h (minimal separation between 24 h, 3 days, and 7 days libraries) (Figure 2A). In contrast, Collection 5 libraries continued to shift progressively toward the lower right region of the PCA plot after 24 h (Figure 2A).

WGCNA using all the genes in the final gene count matrix identified 10 gene co-expression modules, excluding module “gray” containing noisy genes. The number of genes in each module ranged from 307 in module “purple” to 2493 in module “turquoise.” The MEs of all the gene co-expression modules are depicted in Figure S2.

3.4 | Transcriptomic Response to Warm Temperature Exposure

The RNA-seq analyses were organized to separate three sources of transcriptomic variation: the response to warm temperature exposure under forcing conditions, the response associated with field chilling accumulation, and the interaction between field chilling and warm temperature exposure. We first focused on the forcing response by examining gene expression changes

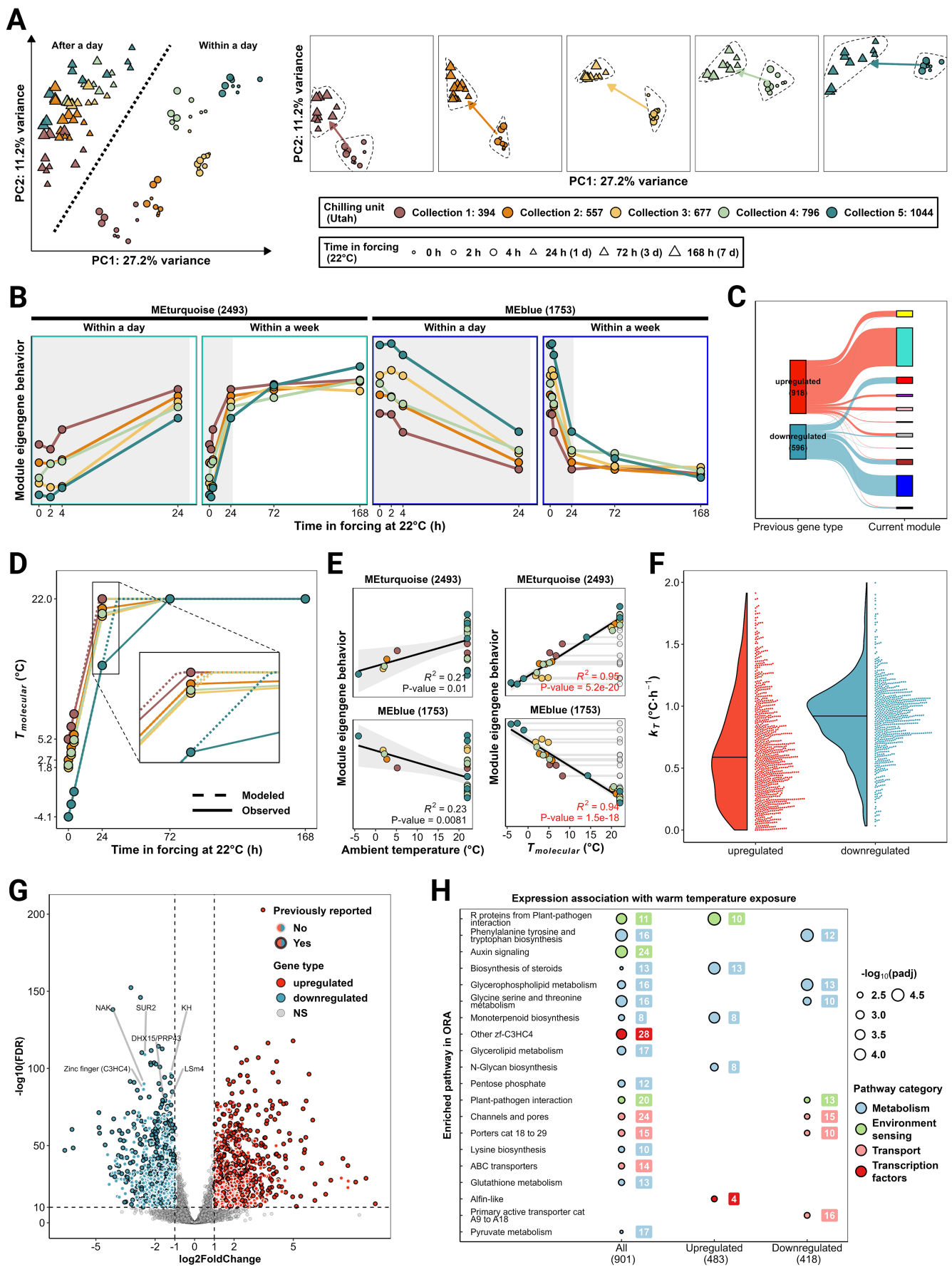


FIGURE 2 | Legend on next page.

FIGURE 2 | Transcriptomic response to warm temperature exposure. (A) PCA of the active transcriptome, showing PC1 and PC2; (B) Expression patterns of the MEs of co-expression modules “turquoise” and “blue,” which only responded to warm temperature exposure; (C) Distribution of previously identified temperature-responsive genes across WGCNA-defined modules; (D) Computed $T_{molecular}$ of each sample, derived from the k_T of module “turquoise”; (E) Correlation between MEs of modules “turquoise” and “blue” with both ambient temperature and $T_{molecular}$; (F) k_T of individual temperature-responsive genes; (G) Volcano plot of differentially expressed temperature-responsive genes; (H) Pathway enrichment analysis of the identified temperature-responsive genes.

during the 22°C deacclimation-forcing assays and then assessed whether these responses differed among collections with different levels of field chilling.

The MEs of the two largest gene co-expression modules “turquoise” and “blue” exhibited distinct dynamic behaviors during the forcing assays. These trends were closely mirrored or inversely correlated with PC1 from PCA (Figure 2B and Figure S3). Together, these two modules represented nearly 40% of the active transcriptome in our study. Although the initial ME expression levels under field conditions differed across collections, a common temporal response pattern emerged across all field collections. Little change was observed within the first 2 h of forcing, but a slight increase/decrease occurred by 4 h. The greatest transcriptomic shift occurred between 4 h and 24 h, and the overall magnitude of change during this period was consistent across all collections, regardless of previous field conditions. After 24 h, samples from Collections 1 through 4 showed minimal additional change in ME expression at 3 and 7 days. In contrast, the ME expression in Collection 5 samples increased/decreased between 24 h and 3 days (Figure 2B).

Temperature-responsive genes identified in a previous study (Wang, Kovaleski, and Londo 2024) were compared with the genes associated with the WGCNA co-expression modules from this study. Among the 918 genes previously identified as positively correlated with temperature, 660 (72%) were assigned to the module “turquoise” (Figure 2C). Among the 596 genes previously identified as negatively correlated with temperature, 357 (60%) were assigned to module “blue.” This suggests that modules “turquoise” and “blue” collectively captured temperature-responsive genes in our dataset.

3.5 | Molecular Temperature Model

Based on the strong overlap between the genes in modules “turquoise” and “blue” and previously identified temperature-responsive genes, we developed an empirical gene expression model to describe transcriptomic adaptation to warm temperature exposure during the forcing assays. The model assumes that when ambient temperature increases to 22°C, gene expression shifts linearly toward a target expression level with a constant k_T , independent of field chilling. Using ME expression values from the “turquoise” and “blue” modules, we estimated k_T through the empirical gene expression model (Equation 1). For the module “turquoise” that represents a positive shift in gene expression toward the 22°C target, the k_T was 0.76°C h⁻¹. For the module “blue” that represents a negative shift in expression, the k_T was 0.77°C h⁻¹. These rates were subsequently used to calculate the $T_{molecular}$, which theoretically aligns with the current expression of the temperature-responsive genes, as a

function of baseline temperature under field conditions, time in forcing, and the target temperature of 22°C (Equations 2 and 3). As calculated, the $T_{molecular}$ plateaued at approximately 23 h for Collection 1, while Collections 2 through 4 reached their target expression levels shortly after 24 h (Figure 2D). In Collection 5, the $T_{molecular}$ continued to change and reached stability around 35 h (Figure 2D). To validate this model, we performed correlation analysis comparing ME expression values of modules “turquoise” and “blue” to both ambient temperature and $T_{molecular}$. Regression using the $T_{molecular}$ yielded significantly improved fit (higher R^2 and lower p -values; Figure 2E).

Using $T_{molecular}$ as the only numeric factor in differential expression analysis, we identified 1820 differentially expressed genes (DEGs) with highly significant expression correlations ($p_{adj} \leq 1e-10$; Data S1). Of these, 1029 were upregulated and 791 were downregulated in response to warm temperature exposure. The k_T of individual upregulated genes ranged from 0°C to 2°C h⁻¹, with an average of 0.66°C h⁻¹ and a distribution skewed toward 0°C to 0.5°C h⁻¹. To compare, the k_T of individual downregulated genes followed a normal distribution within the same range, with a higher average of 0.96°C h⁻¹ (Figure 2F).

Many of the most significantly differentially expressed genes with the greatest log₂ fold changes overlapped with the previously identified temperature-responsive genes (Wang, Kovaleski, and Londo 2024) (Figure 2G). However, among the downregulated genes, we identified several novel and highly significant members that had not been previously reported. These include *VIT_01s0026g02310* (encoding sphinganine C4-monooxygenase/SUR2), *VIT_14s0108g01040* (KH domain-containing protein), *VIT_02s0012g00150* (Numb-associated kinase/NAK), *VIT_17s0000g06610* (Zinc finger protein C3HC4), *VIT_07s0005g01770* (DEAH-box RNA helicase/DHX15), and *VIT_07s0005g00230* (U6 snRNA-associated Sm-like protein/LSm4). Pathway enrichment analysis of the temperature-responsive genes revealed 20 significantly enriched pathways primarily related to the Metabolism and Environmental Sensing categories (Figure 2H).

3.6 | Transcriptomic Response to Chilling

The second analysis aimed at deciphering the impact of field chilling accumulation on gene expression during the forcing assays. We identified several co-expression modules whose MEs showed expression patterns that either correlated with field chilling or reflected the interactions of chilling and forcing exposure.

We identified one co-expression module “black” that was negatively correlated with chilling accumulation but exhibited no

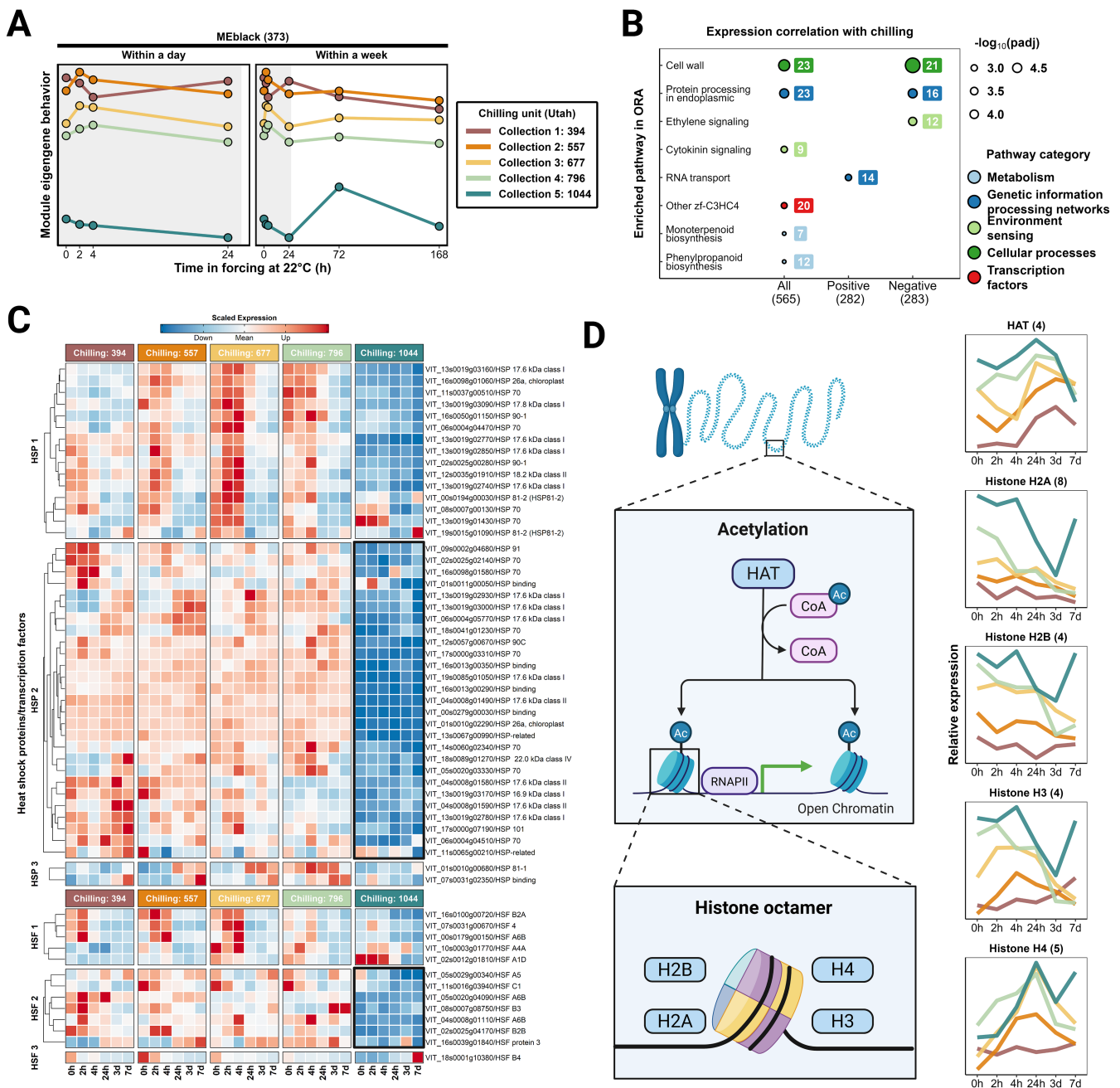


FIGURE 3 | Transcriptomic response to chilling. (A) Expression pattern of the ME of module “black” that showed chilling only response; (B) Pathway enrichment analysis of the chilling-responsive genes; (C) Expression of all genes encoding HSPs and HSFs; (D) Schematic of histone acetylation and the summed expression of genes encoding histone octamer components (H2A, H2B, H3, and H4) and HATs. Numbers beside protein names indicate the number of gene copies included in the expression sum. The figure was assembled in and the illustrations were obtained from BioRender (<https://www.biorender.com/>).

interaction effects during deacclimation assays (Figure 3A). Pathway enrichment analysis of the identified 565 annotated DEGs that showed this behavior (out of a total of 1138 DEGs; Data S2) revealed eight significantly enriched pathways (Figure 3B). In the protein processing in the endoplasmic reticulum pathway, 14 out of 16 DEGs were negatively correlated with chilling and were associated with heat shock proteins (HSPs). Further examination of all 1138 chilling-responsive DEGs expanded the list to include 21 DEGs that are associated with HSPs or heat shock transcription factors (HSFs). The expression of these genes showed a significant and consistent

negative correlation with previous chilling (Figure S4). Given the large number of DEGs associated with HSPs and HSFs, we examined the expression pattern for all genes encoding HSPs and HSFs in the current transcriptome annotation. Nearly two-thirds of all HSP genes followed the same expression behavior as the ME of module “black” (Figure 3C).

A second enriched group of genes correlated with field chilling was histone-related proteins. These 13 DEGs encode H2A (2 DEGs), H2B (3 DEGs), H3 (3 DEGs), and H4 (5 DEGs), and their expression pattern was the inverse pattern of the ME of

module “black” (Figure S5). Using the same approach, we investigated the expression patterns of all genes encoding histone-related proteins and found that the summed expression of all the genes encoding for the proteins in histone octamer: H2A (eight copies), H2B (four copies), H3 (four copies), and H4 (five copies), and histone acetyltransferase (HAT, four copies) also followed this pattern (Figure 3D).

3.7 | Transcriptomic Response to the Interaction of Chilling and Warm Temperature Exposure

Next, we evaluated the impact of the interaction of warm temperature exposure and field chilling accumulation on gene expression dynamics under forcing conditions. We observed that two co-expression modules, “brown” and “yellow,” were correlated with chilling accumulation and subsequently shifted during the forcing assays. For module “brown,” ME expression was positively correlated with chilling under field conditions, remained stable during the first 24 h of forcing, and then progressively decreased in higher-chilling samples while slightly increasing in lower-chilling samples, resulting in convergence by 7 days. In contrast, module “yellow” was negatively correlated with chilling under field conditions but exhibited the opposite trend during forcing: ME expression increased in high-chilling samples and decreased in low-chilling ones, also converging by 7 days. These dynamics reveal chilling-dependent patterns in response to warm temperature, which we describe as chilling-positive convergence (module “brown”) and chilling-negative convergence (module “yellow”), both characterized by initial chilling-associated divergence followed by convergence during the forcing assay (Figure 4A).

We also observed that the behavior of these MEs (chilling-positive convergence and chilling-negative convergence) appeared to be closely associated with bud cold hardiness. Using the observed k_{deacc} and the initial cold hardiness values for each collection, we calculated cold hardiness at each timepoint during the five deacclimation assays (Equation 4, Figure 4B). The ME expression patterns of modules “brown” and “yellow” exhibited strong and statistically significant correlations with the calculated cold hardiness values across timepoints (Figure 4C).

Pathway enrichment analysis of the annotated DEGs that showed the ME patterns of modules “brown” and “yellow” (Data S3) identified six overrepresented pathways (Figure 4D). Among the 17 DEGs associated with the overrepresented cell wall pathway, 15 exhibited chilling-negative convergence behavior. We then examined the overall expression patterns of all cell wall-related genes that passed low-count filtering. This analysis revealed that, among 33 annotated proteins in the cell wall pathway, 13 exhibited an expression pattern that matched either the chilling-negative convergence pattern or the chilling-only response pattern. The total expression of the genes encoding these proteins was higher under field conditions in early collections (lower chilling), followed by a gradual convergence toward similar expression levels across all collections as the forcing assay progressed (Figure 4E).

In addition, although not statistically overrepresented in the pathway enrichment analysis, several genes involved in ABA and auxin metabolism exhibited expression patterns associated with the interaction between warm temperature exposure and chilling. In the ABA metabolism pathway, the summed expression of 9-cis-epoxycarotenoid dioxygenase (NCED, key ABA biosynthesis enzyme) genes was highest in early collections under field conditions but declined and eventually converged to a minimum level across all collections by 7 days. Conversely, the summed expression of ABA 8'-hydroxylase (ABA 8'OH, key ABA degradation enzyme) genes was highest in later collections under field conditions and similarly decreased over time, converging by 7 days (Figure 4F). In the auxin metabolism pathway, the summed expression of Gretchen Hagen 3 (GH3, key auxin conjugation enzyme) genes and amidase genes (key auxin de-conjugation enzyme) followed the chilling-negative convergence pattern and chilling-positive convergence pattern, respectively (Figure 4G).

3.8 | Plant Hormone Signaling Pathways and ICE, CBF/DREB1 Genes

We manually examined the expression of the genes involved in several plant hormone signaling pathways that have been previously reported to be correlated with grapevine cold hardiness and deacclimation (Chai et al. 2019; Londo and Kovaleski 2019; Kovaleski and Londo 2019; Wang, Kovaleski, and Londo 2024) and the members of the ICE and CBF/DREB1 genes (Xiao et al. 2006; Noriega et al. 2024).

In the ethylene signaling pathway, the overall expression of signaling components did not show a consistent response to chilling. However, several members, including Ethylene Insensitive 3 (EIN3), Mitogen-Activated Protein Kinase 3 (MPK3), Constitutive Triple Response 1 (CTR1), and EIN3-Binding F-box Protein 1 (EBF1), responded rapidly to warm temperature exposure under the forcing assays. We also identified six ethylene responsive transcription factors (AP2/ERFs) that exhibited a negative correlation with field chilling (Figure S6), which aligns with the overrepresentation of ethylene signaling among the DEGs that showed the chilling only response (Figure 3B). In the ABA signaling pathway, the overall expression of the genes encoding Protein Phosphatase 2C (PP2C, eight genes) showed a general chilling-only response, being negatively correlated with field chilling. The overall expression of the genes encoding SNF1-Related Protein Kinase 2 (SnRK2, four genes) showed a chilling-positive convergence pattern (Figure S7). In the auxin signaling pathway, the overall expression of multiple auxin-responsive transcription factors (ARFs) showed a clear response to warm temperature exposure, though in both directions: some increased and others decreased under forcing conditions. Similarly, components of auxin receptors and signaling intermediates exhibited rapid but divergent responses to warm temperature exposure (Figure S8). Only three genes in the ICE and CBF/DREB1 gene group passed the low count filtering. The expression of the two DREB genes, *VIT_11s0016g02140* and *VIT_13s0067g01960*, displayed a general chilling-positive convergence behavior (Figure S9).

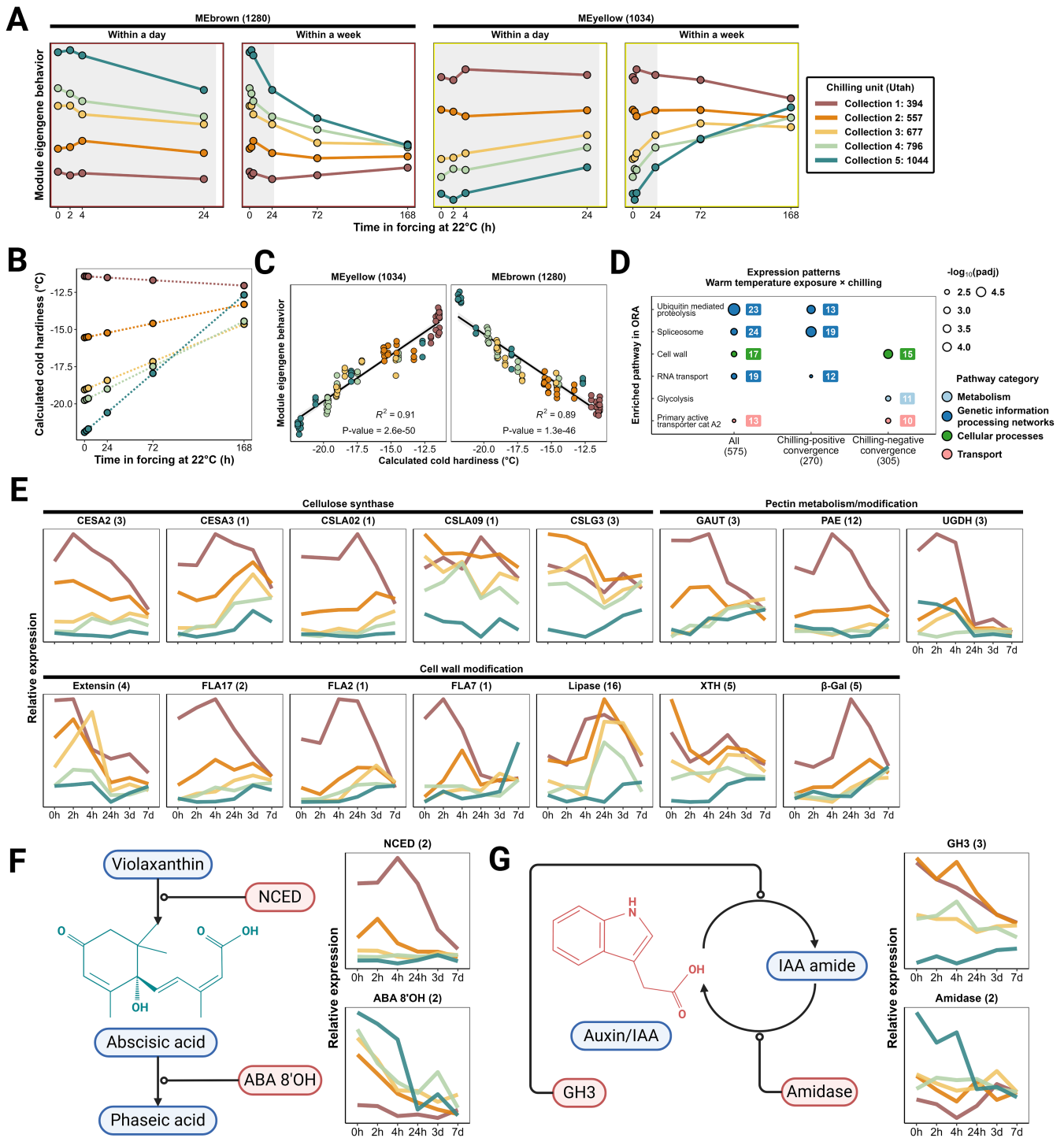


FIGURE 4 | Transcriptomic response to the interaction of chilling and warm temperature exposure. (A) Expression patterns of MEs for modules “brown” and “yellow,” showing “chilling-positive convergence” and “chilling-negative convergence” patterns, respectively; (B) Calculated cold hardness of buds during the experiment; (C) Correlation of MEs from modules “brown” and “yellow” with calculated cold hardness; (D) Pathway enrichment analysis of the chilling-positive convergence and chilling-negative convergence genes; (E) Summed expression of genes encoding cell wall proteins that followed the chilling-negative convergence pattern; (F) Schematic of abbreviated ABA metabolism and the summed expression of NCED and ABA 8’OH genes; (G) Schematic of auxin-amino acid conjugation and deconjugation, and the summed expression of GH3 and amidase gene. Numbers beside protein names indicate the number of gene copies included in the expression sum.

4 | Discussion

Warming and increasingly unstable winters driven by climate change may threaten the sustainability of global grape and wine production by disrupting the physiological progression

of grapevine dormancy. Although the physiological and transcriptomic mechanisms underlying cold acclimation and deacclimation have been a topic of recent studies (Chai et al. 2019; Kovaleski and Londo 2019; Camargo-Alvarez et al. 2020; North and Kovaleski 2023; Wang, Kovaleski, and Londo 2024; Wang,

Moghe, et al. 2024; Wang et al. 2025; Noriega et al. 2024; Londo and Kovaleski 2025; De Rosa et al. 2025; Guo et al. 2025), far less is known about how field chilling accumulation shapes transcriptomic responsiveness during the dormant season. This study aimed to uncover how changing chilling accumulation influences gene expression responses of buds during deacclimation in forcing conditions.

In this study, “Cabernet Sauvignon” grapevine buds were sampled from the field five times (Collection 1 to 5) during the dormant season. The chilling accumulation of the five collections ranged between 394 and 1044 chilling hours (Utah model). Based on the previously established relationship between chilling accumulation and deacclimation potential in “Cabernet Sauvignon” (Londo and Kovaleski 2025), buds were estimated to have reached approximately 0%, 12%, 25%, 40%, and 75% of their maximum deacclimation potential in Collections 1 through 5, respectively. Consistent with these estimates, the k_{deacc} increased proportionally during the deacclimation assays, further supporting the chilling-mediated progression of deacclimation potential (Londo and Kovaleski 2025). Based on the chilling requirement of “Cabernet Sauvignon,” buds in Collection 1 were endodormant, transitioning from endodormancy to ecodormancy in Collection 2 through 4, and ecodormant in Collection 5 (Londo and Johnson 2014; Kovaleski et al. 2018; Londo and Kovaleski 2025).

The gene expression analysis conducted here revealed new insights into the regulation of chilling, warm temperature sensing, and their interaction under growth-permissive conditions where deacclimation rate varies. The majority of genes from the four largest modules in this study, “turquoise,” “blue,” “brown,” and “yellow,” were broadly classified as temperature-responsive in a previous study using weekly RNA-seq sampling (Wang, Kovaleski, and Londo 2024). However, the modules “turquoise” and “blue” clearly displayed a much faster transcriptomic response to the warm temperature exposure, with most changes occurring within the first 24 h across all five field collections, regardless of field chilling accumulation. In contrast, modules “brown” and “yellow” showed a slower mode of action, where both the rate and the direction of expression change were impacted by previous chilling, resulting in a delayed expression convergence by 7 days. This clear differentiation enables the separation of temperature-responsive genes (genes in modules “turquoise” and “blue”) from those genes with responses modulated by chilling (genes in modules “brown” and “yellow”). Although several studies have explored transcriptomic changes during grapevine deacclimation (Kovaleski and Londo 2019; Wang et al. 2022; Noriega et al. 2024; De Rosa et al. 2025), this is the first study to incorporate field chilling accumulation as a factor in the analysis. This approach enabled the identification of genes that respond specifically to chilling, independent of warm temperature exposure, such as those in module “black.”

To further explore the expression pattern of the temperature-responsive genes, we conducted empirical modeling and differential expression analysis based on computed $T_{molecular}$ for genes that respond rapidly to warm temperature exposure. Analysis of module eigengene (ME) behavior revealed three key insights: (1) minimal transcriptomic changes occurred within the first 4 h; (2) most gene expression changes occurred between 4 and 24 h

after warming; and (3) the magnitude and rate of transcriptomic change were largely independent of field sampling time, indicating a similar response to warming across different field chilling accumulation. Notably, the maximum temperature increase in this experiment was 26°C (from the coldest field sample to the growth chamber condition), and most of the gene expression changes were completed within 1 day. This suggests that in dormant grapevine buds, simple temperature-responsive gene regulation is rapid and chilling-independent.

Based on this behavior, we calculated a theoretical molecular temperature response rate, termed k_T , as $0.76^{\circ}\text{C h}^{-1}$ for up-regulated and $0.77^{\circ}\text{C h}^{-1}$ for downregulated gene modules, respectively. These rates suggest a maximum transcriptomic adjustment capacity of approximately 18°C per day. Therefore, under natural conditions, when ambient temperatures fluctuate more than 18°C within a single day, the transcriptomic machinery in dormant grapevine buds may be unable to fully adjust. Since transcriptional regulation often represents the earliest molecular response to environmental stimuli (Nievola et al. 2017), such mismatches between diurnal temperature shifts and transcriptomic responsiveness may lead to insufficient physiological responses, reducing the plant's ability to adjust to environmental stress in a timely manner. As rapid and extreme temperature fluctuations are projected to become more frequent and intense under future climate scenarios (Wu et al. 2025), further research is urgently needed to evaluate how large winter temperature swings disrupt the physiological processes in the dormant season.

Using the empirically determined k_T , we also calculated the $T_{molecular}$ for each RNA-seq library, which served as a proxy variable for identifying temperature-responsive genes. A total of 1820 DEGs exhibited highly significant correlations with $T_{molecular}$, including 1029 upregulated and 791 downregulated in response to warm temperature exposure. The abundance of highly significant temperature-responsive genes identified in this study indicates that a substantial portion of the active transcriptome of grapevine buds in the dormant season is associated with temperature sensing and response, and that this regulatory mechanism remains active regardless of the dormancy status. While the overall behavior aligned with the eigengene patterns of modules “turquoise” and “blue,” we observed a slight difference in k_T at the individual gene level. Notably, genes that were downregulated in response to warm temperature exposure had, on average, a 31% higher k_T compared to the upregulated genes. If this relationship holds true under natural conditions, it suggests that during a cold event, cold-induced genes are upregulated more rapidly than cold-suppressed genes are downregulated. This asymmetric response may reflect underlying biological mechanisms, where the suppression of gene expression under cold may be partially passive (e.g., reduced enzymatic activity), whereas the induction of cold-responsive genes involves a faster active transcriptional programming (Ding et al. 2024).

Most of the temperature-responsive genes identified in this study have been reported in a previous study (Wang, Kovaleski, and Londo 2024), but we also identified six highly significant new members that were downregulated under warm temperature exposure. With the exception of *VIT_02s0012g00150* (NAK gene), whose function in plants

remains poorly characterized, the other five genes have been implicated in key biological processes in other plant species. *VIT_01s0026g02310* encodes SUR2, a key enzyme in sphingolipid biosynthesis, a pathway implicated in cold stress signaling and membrane stabilization (Ali et al. 2018). *VIT_14s0108g01040* encodes a KH domain-containing RNA-binding protein, which has been associated with thermotolerance and heat-induced gene regulation through modulation of stress-responsive transcription factors (Guan et al. 2013). *VIT_07s0005g01770* encodes DHX15, and *VIT_07s0005g00230* encodes LSM4. These proteins are involved in RNA modification, and their downregulation may mark reduced demand for translational regulation under warming (Bouveret et al. 2000; Mosallanejad et al. 2014; Kerbler and Wigge 2023). *VIT_17s0000g06610* encodes a C3HC4-type RING zinc finger protein/E3 ligase and is likely involved in protein ubiquitination and degradation during abiotic stress. E3 ligases have been shown to regulate abiotic stress adaptation in different crops (Agarwal and Khurana 2020; Han et al. 2021). Detailed functional characterization of these genes in grapevine is still needed, and their significant downregulation under warm temperature exposure suggests that they may play conserved roles in maintaining transcriptomic stability and dormancy-associated signaling.

Pathway enrichment analysis revealed that temperature-responsive genes are predominantly associated with metabolism and environmental sensing pathways, as seen in a previous study (Wang, Kovaleski, and Londo 2024). These two pathway groups likely act synergistically: temperature sensing initiates metabolic reprogramming, while the resulting metabolic shifts can, in turn, modulate and reinforce stress signaling. Among all plant hormones, auxin was the only signaling pathway significantly overrepresented among the temperature-responsive genes. Detailed investigation revealed that eight ARFs, two auxin receptors, four negative regulators, and four positive regulators of auxin signaling exhibited expression patterns that were highly correlated with the $T_{molecular}$. Although the direction of regulation varied (some were upregulated while others were downregulated under warm temperature exposure), this overall pattern suggests a critical role for auxin signaling in immediate temperature sensing and response, which is consistent with previous findings (Gray et al. 1998; Bellstaedt et al. 2019). The rapid modulation of auxin signaling likely facilitates swift growth adjustments, such as thermomorphogenesis, and integrates with other regulatory networks to coordinate adaptation and the initiation of developmental transitions under changing thermal conditions (Bianchimano et al. 2023).

Another key insight from this study is the identification of three gene co-expression modules (“black,” “brown,” and “yellow”) that showed significant associations with chilling accumulation. The module associated with responses correlated with chilling alone (module “black”) included the cell wall pathway, protein processing in the endoplasmic reticulum, and ethylene signaling. The cell wall pathway was also significantly enriched among genes responsive to the interaction between warm temperature exposure and chilling (modules “brown” and “yellow”), with nearly 40% of the pathway’s proteins following either a chilling-only (negative) pattern or a chilling-negative convergence pattern. Cell wall remodeling has long been recognized as

a key mechanism in chilling-mediated dormancy progression. For example, in apple (*Malus domestica*), DNA methylation regulates genes like *Pectinesterase 3*, which promotes pectin remodeling critical for cell wall flexibility during chilling-induced dormancy release (Kumar et al. 2016). In *Polygonatum kingianum*, chilling has been shown to cause structural changes in plasmodesmata and the cell wall, supported by transcriptomic evidence for the upregulation of cell wall-loosening genes (Wang et al. 2019). Similarly, in *Arabidopsis thaliana*, chilling and sub-zero acclimation trigger broad alterations in cell wall composition, including the induction of enzymes such as pectin methylesterases and xyloglucan endotransglucosylase/hydrolases like XTH19, which are important for structural remodeling (Takahashi et al. 2019, 2021). In grapevines, another group of genes encoding cell wall proteins and enzymes, such as cellulase, pectate lyase, pectin-acetyltransferase and polygalacturonase, was found intensively upregulated along with other growth-related pathways when approaching budbreak (Wang et al. 2025). These changes are believed to enhance intercellular communication and prime tissues for growth reactivation post-dormancy (Alonso Baez and Bacete 2023). Our findings suggest that intensive cell wall remodeling occurs during the early dormant season in grapevine buds, and this remodeling diminished as chilling accumulates. Warm temperature exposure further reduces the intensity of this process. This observation suggests that unseasonal warming events in early winter may suppress cell wall remodeling activity, potentially disrupting normal dormancy cycling (Kovaleski 2024).

Genes encoding HSPs and HSFs exhibited expression patterns negatively associated with chilling and yet remained unresponsive to warm temperature exposure. Notably, the majority of these HSPs were small HSPs, which function as molecular chaperones, stabilizing and refolding proteins during stress conditions (Bourguine and Guihur 2021). Although these proteins have been repeatedly reported for their expression correlation and stabilizing functionality under cold stress in various plant species (Takemura and Tamura 2016; ul Haq et al. 2019; Chang et al. 2021; Sadura and Janeczko 2024), limited information is available for their potential function in the chilling-mediated dormancy progression. The genes encoding several histone-related proteins, including H2A, H2B, H3, H4, and HATs, were also found to be positively correlated with chilling. The upregulation of the genes encoding these proteins during the dormant season suggests an active chromatin remodeling process in grapevine buds. Similar histone modification has been found in poplar (Conde et al. 2019), aspen (Karlberg et al. 2010), chestnut (Santamaría et al. 2009), apple (Kumar et al. 2016), and grapevine (De Rosa et al. 2024), and has been proposed to be a key epigenetic mechanism underlying the chilling-mediated dormancy transition by progressively enabling the onset of growth-related genes.

We found that the two modules that revealed the interaction between warm temperature exposure and chilling, “brown” and “yellow,” were highly correlated with cold hardiness. This result suggests that a subset of the active transcriptome in dormant season grapevine buds integrates both chilling and immediate thermal cues to reflect their physiological state. In ABA metabolism, NCED (biosynthesis) genes decreased and ABA 8’OH (degradation) genes increased with chilling, suggesting

a decline in ABA content, consistent with previous findings (Zheng et al. 2015; Zheng, Acheampong, et al. 2018). Under forcing conditions, expression of both enzyme types dropped to minimal levels regardless of chilling history. Similarly, GH3 (auxin conjugation) decreased while amidase (deconjugation) increased with chilling, indicating a rise in free auxin, followed by convergence under forcing. Several ERF genes in the ethylene signaling pathway also showed this pattern. Our analysis indicates, first, that chilling accumulation drives transcriptomic shifts across multiple hormone pathways, potentially reflecting a gradual, hormone-mediated dormancy progression. Second, warm temperature exposure appears to override this chilling-dependent transcriptional regulation, which potentially resets the hormone-associated gene expression to a uniform level and

turns transcriptomic switch from dormancy maintenance to growth reactivation.

Together, our findings provide a comprehensive examination of the transcriptomic dynamics underlying grapevine bud dormancy transitions and deacclimation. Based on the expression patterns observed in this study and existing knowledge of dormant season physiology, we propose a theoretical model to illustrate how chilling accumulation and warm temperature exposure regulate transcriptomic behavior in dormant buds (Figure 5). A large portion of the transcriptome is rapidly responsive to warm temperatures in a chilling-independent manner, which potentially indicates a conserved and constrained temperature-sensing mechanism throughout the dormant season. These

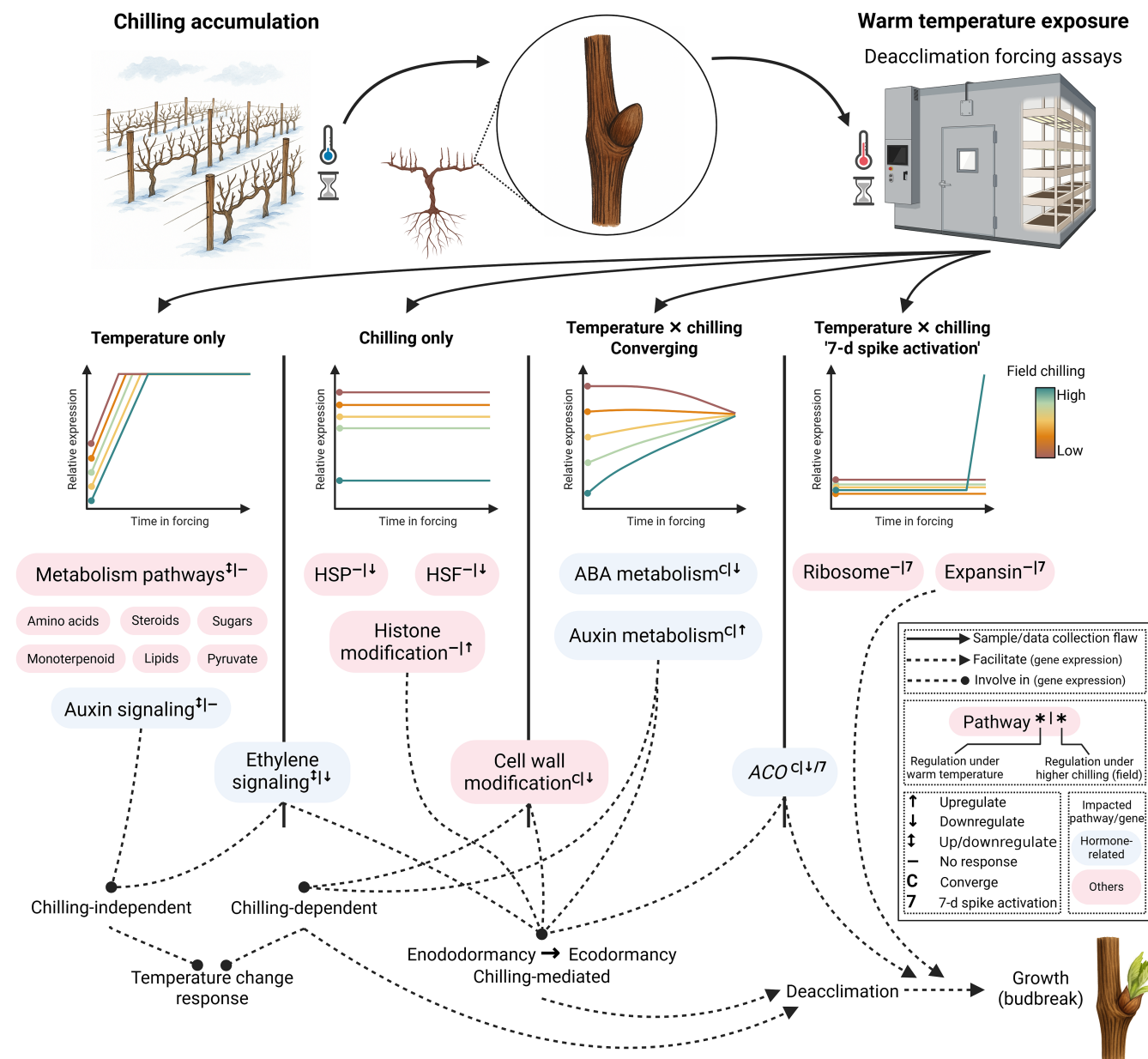


FIGURE 5 | Conceptual model of chilling and warm temperature interaction in grapevine bud transcriptomic regulation. Schematic representation summarizing the roles of chilling accumulation and warm temperature exposure in regulating transcriptome during dormancy progression, deacclimation, and growth reactivation in grapevine buds. Refer to Figure 1A for the experimental workflow linking field collection dates, field chilling accumulation, and subsequent lab-based deacclimation forcing intervals. The figure was made in and the illustrations were obtained from BioRender (<https://www.biorender.com/>).

genes, largely involved in pathways of metabolism, environmental sensing, and auxin signaling, potentially form the core of the immediate temperature-sensing and response mechanism. In contrast, chilling modulates the dormant-season transcriptome in two distinct ways. First, chilling-only responsive genes, such as those involved in histone modification and HSP synthesis, continuously change the internal bud state under field conditions to facilitate dormancy progression. Second, genes responsive to the interaction of chilling and warm temperature exposure, particularly those involved in ABA and auxin metabolism and cell wall remodeling, could have a dual role in both dormancy progression in the field and chilling-dependent deacclimation/growth under growth-permissive conditions.

Overall, the temperature-dependent, chilling-dependent, and chilling-temperature interaction gene expression behaviors observed in this study indicate the complex regulatory framework underlying grapevine dormant season physiology. By disentangling the effects of chilling and warm temperature exposure, we were able to uncover the transcriptomic basis of temperature sensing and the chilling-mediated dormancy transitions in grapevine. Several important questions remain. First, activation of the temperature-responsive genes identified in this study does not appear sufficient to induce physiological changes in cold hardiness (Wang, Kovaleski, and Londo 2024). How these genes contribute to preparing the plant for temperature adaptation, either independently or dependently on dormancy transitions, remains an open question for future research. Second, our experimental design assumes that chilling is the driver of dormancy transition. However, because chilling accumulation is inherently confounded with time, and given ongoing debates about the accuracy of chilling models (including the Utah model used in this study) (Lin et al. 2022; North et al. 2024; Wang, Bai, et al. 2024), we cannot exclude the possibility that the observed gene expression changes are driven by chronological time rather than chilling per se. Future studies involving large-scale, high-resolution field RNA-seq datasets will be critical to distinguish chilling effects from temporal effects and to validate the concept of chilling-responsive transcriptomic regulation. Third, a substantial proportion of the genes showing a significant correlation with chilling are unannotated, limiting our ability to interpret their functional relevance. Comprehensive annotation of these genes, potentially through comparative genomics and functional characterization, will be essential to fully uncover the underlying biology. Lastly, multi-omics validation (e.g., proteomics, metabolomics, and epigenomics) will be required to confirm the roles of candidate genes and to validate the findings from this study, which will help build a more mechanistic understanding of how temperature changes and chilling accumulation shape dormancy regulation in grapevines.

Author Contributions

J.P.L. and H.W. designed and conducted the experiment. H.W. collected samples, conducted deacclimation assays, extracted RNA, and conducted the RNA-seq analysis. J.P.L. and H.W. wrote the manuscript.

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Disclosure

No generative AI tools have been used for this study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The RNA-seq raw data along with the processed gene count matrix and sample metadata supporting the conclusions of this article are available in the NCBI-GEO repository accession GSE299692.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Identification of transcriptome outliers using Principal Component Analysis (PCA). **Figure S2:** Expression pattern of the module eigengene (ME) of each gene co-expression module identified by weight gene co-expression network analysis (WGCNA). **Figure S3:** Correlation analysis of the MEs of gene co-expression modules "turquoise" and "blue" with PC1 from PCA. **Figure S4:** Expression of the heat shock protein (HSP) and heat shock transcription factor (HSF) genes significantly correlated with chilling. For clarity, expression values were summed for genes encoding the same protein. Numbers beside protein names indicate the number of gene copies included in each sum. **Figure S5:** Expression of the histone octamer protein (H2A, H2B, H3 and H4) genes significantly correlated with chilling. For clarity, expression values were summed for genes encoding the same protein. Numbers beside protein names indicate the number of gene copies included in each sum. **Figure S6:** Expression of all the genes encoding ethylene signaling components that showed response to warm temperature exposure, chilling, or their interaction. For clarity, expression values were summed for genes encoding the same protein. Numbers beside protein names indicate the number of gene copies included in each sum. **Figure S7:** Expression of all the genes encoding three major ABA signaling components. For clarity, expression values were summed for genes encoding the same protein. Numbers beside protein names indicate the number of gene copies included in each sum. **Figure S8:** Expression of all the genes encoding auxin signaling components that showed response to warm temperature exposure, chilling, or their interaction. For clarity, expression values were summed for genes encoding the same protein. Numbers beside protein names indicate the number of gene copies included in each sum. **Figure S9:** Expression of all genes with detectable expression encoding ICE or CBF/DREB. **Data S1:** Statistics of the differential expression analysis and detailed information of the genes that showed significant correlation with molecular temperature. **Data S2:** Statistics of the differential expression analysis and detailed information of the genes that showed significant correlation with chilling. **Data S3:** Statistics of the differential expression analysis and detailed information of the genes that showed chilling-negative convergence and chilling-positive convergence expression patterns.